Austin Luk

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EDUCATION

University Of British Columbia — ***Graduating May 2027***

Bachelor of Science in Computer Science

Relevant Coursework: Machine Learning (CPSC 330), Introduction to Artificial Intelligence (CPSC 322) Data Structures and Algorithms (CPSC 221), Software Construction (CPSC 210), Database Systems (CPSC 304),

Langara College — ***Sept 2023 – April 2025***

Associate of Science Computer Science

Activities and societies: Vice–President of Langara Volleyball Club, Langara Computer Science Club (LCC)

EXPERIENCE

Code Ninjas — **March 2025 - Current**

Instructor — ***Vancouver, BC***

• Troubleshot software, hardware, and connectivity issues in a fast-paced classroom environment, supporting 50+ students.

• Diagnosed and resolved technical problems across Windows and browser-based platforms under time constraints

• Taught programming fundamentals (Scratch, JavaScript, Lua, C#/Unity) by guiding students through debugging and problem-solving workflows

Headstarter AI — **July 2024 – Sept 2024**

Software Engineer Fellow — ***Vancouver, BC***

• Debugged and fixed environment and dependency issues across full-stack applications, improving development.

• Collaborated in agile teams to build and deploy full-stack applications using MVC architecture (Next.js + backend APIs), contributing features across frontend and backend

• Designed AI-powered applications using **OpenAI APIs + vector-based retrieval (RAG)** to improve response relevance.

PROJECTS

CourtIQ | Python, PyTorch, OpenCV

• Developed a real-time computer vision system to track volleyball player movements and ball trajectory from video input

• Trained and **fine-tuned** object detection models in PyTorch for motion tracking

• Built a data processing pipeline to extract positional data and generate gameplay analytics (e.g., player movement, ball speed, rally patterns)

• Developed a data pipeline to extract positional data and compute metrics (player movement paths, ball speed, rally duration)

TaPRO | Gemini API, Next.js, Node.js, Supabase, AWS

• Built an AI-powered virtual teaching assistant platform to support university office hours for students and professors.

• Designed data pipelines to store assignments, course materials, and interaction summaries in Supabase for fast retrieval and reuse.

• Implemented optimized retrieval pipelines to reduce API payload size and improve response latency by 40%.

ThenStep| Next.js, Bun, OpenStreetMap, Google Maps API, Puppeteer

• Developed an advanced map application that allows users to search for both normal driving and walking routes, as well as scenic alternatives.

• Used Puppeteer for web scraping to gather data on safety reports, local meetups, and community events.

• Integrated multiple APIs **(OpenStreetMap + Google Maps)** for dynamic route rendering and interactive mapping.

TECHNICAL SKILLS

Operating Systems: macOS, Windows, Linux

Languages: Python, Java, JavaScript, TypeScript, C++, SQL, HTML, CSS,

Frameworks & Libraries: React, Node, Express, REST APIs, Git, MySQL, MS SQL, MongoDB

DevOps: Amazon Web Services (AWS), Docker, Kubernetes, Azure, CI/CD

Certifications: **AWS Certified Cloud Practitioner** (In Progress – Expected May 2026)